01 / Chapter map and tutorial relevance

BBMS2006-3-VQ matching and control of breathing-23Sept2026.pdf | 21 source pages

Two connected problems

Matching: does each alveolus receive appropriate amounts of air and blood?
Control: how do sensors, brainstem circuits and respiratory muscles regulate ventilation?

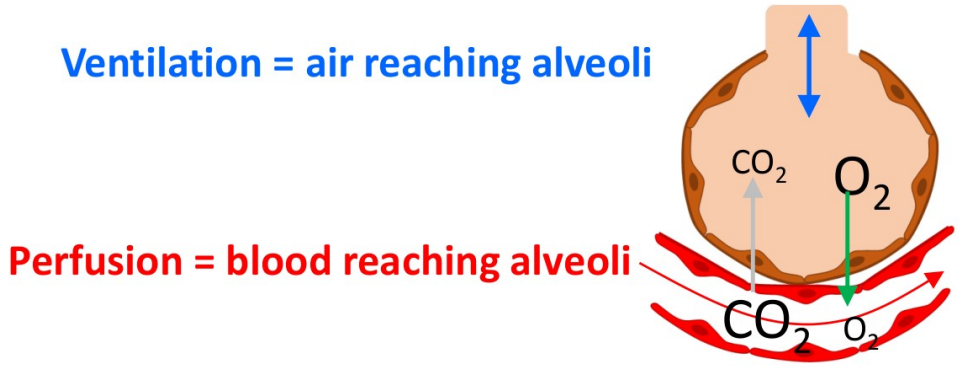
Original tutorial question	Relevance to this chapter	Notes page
Q1 | Compliance always makes breathing easier	Partial overlap: regional compliance is explained here; whole-lung mechanics belongs elsewhere.	3
Q2 | Apex versus base: capillary blood PO_2	Direct: gravity changes V and Q unequally.	5
Q3 | Normal saturation means normal delivery	Lecture 2; supporting distinction for content-weighted blood mixing.	7: link only
Q4 | Altitude, hyperventilation and acidosis	Cross-topic: hypoxic drive versus respiratory alkalosis.	12
Q5 | Right shift reduces pulmonary uptake	Lecture 2; not re-answered here.	—
Q6 | Hypoxemia versus tissue hypoxia	Lecture 2; explains why low Hb is not the same as low PaO_2.	11: link only
Q7 | Hyperventilation at unchanged FiO_2	Cross-topic application of alveolar ventilation.	8
Q8 | Small bronchioles have highest resistance	Airway mechanics; do not confuse with pulmonary vascular resistance.	—
Q9 | Surfactant increases inspiratory work	Other lecture: surface tension and lung mechanics.	—
Q10 | CO decreases Hb–O_2 affinity	Lecture 2; not re-answered here.	—
Q11 | Chronic hypercapnia and ventilatory drive	Direct topic, with an adaptation mechanism supplied as an extension.	10
Q12 | CA inhibitor and bicarbonate excretion	Cross-topic: renal bicarbonate loss and peripheral pH sensing.	12
Q13 | Diving with unchanged nitrogen fraction	Lecture 2: Dalton and Henry.	—
Q14 | Equal minute ventilation means equal VA	Application; dead-space calculation supplied as an extension.	8

Tutorial Q# preserves the original question wording/numbering from BBMS 2006 Tutorial.pdf, p. 1. Answers are reasoned suggestions, not an official key. Lecture references use actual PDF pages. [Extension] marks supporting material not fully developed in the slides; learning-outcome supplements are explicitly labelled. Original teaching images, labels and visible annotations are retained.

02 / V/Q: what is being matched?

Lecture PDF pp. 4, 13–14

Symbol / term	Plain meaning	Representative whole-lung value
V or $\dot{V}A$: alveolar ventilation	Fresh air reaching the gas-exchanging alveoli per minute.	About 4 L/min in the slide example.
Q or $\dot{Q}$: perfusion	Blood flow through pulmonary gas-exchange units per minute.	About 5 L/min.
V/Q ratio	Air supply relative to blood supply; a ratio of rates.	4/5 = 0.8, not 4 or 5.



Original air/blood exchange illustration, lecture p. 4.

Unit state	What is missing?	Limiting V/Q behaviour
Shunt-like	Ventilation is inadequate relative to perfusion.	V/Q ↓; true unventilated perfusion gives V/Q = 0.
Dead-space-like	Perfusion is inadequate relative to ventilation.	V/Q ↑; no perfusion with ventilation gives V/Q → infinity.

Do not turn a whole-lung average into a rule for every alveolus

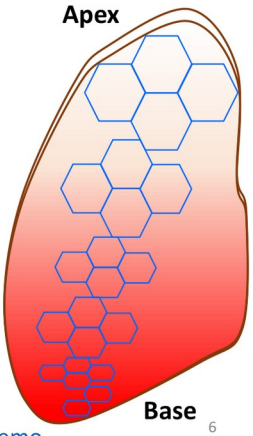
The slides use 0.8 as a resting example and about 0.8–1.2 as a broad whole-lung range. Regional ratios vary. Both gravity and disease can create mismatch. Low V/Q is not necessarily a complete shunt; high V/Q is not necessarily complete dead space.

Notation throughout: PAO_2 / $PACO_2$ = alveolar gas pressures; PaO_2 / $PaCO_2$ = arterial blood pressures; PvO_2 / $PvCO_2$ = mixed venous pressures. Capital A and lower-case a are different compartments.

03 / Gravity changes regional ventilation

Lecture PDF p. 6 | Upright lung at the end of a normal expiration

- Concept: in an upright lung, alveoli at the apex are more expanded at the end of expiration
  - Because gravity pulls visceral pleura at the apex downwards
    - More vacuum at the apex, intrapleural pressure more negative
  - It is harder to expand an 'already-expanded' alveoli (less compliant)
    - During inspiration, alveoli at the apex/base expand more
      - Receive more airflow, more gas exchange



Demo

Original upright-lung explanation and the selected "base" answer, lecture p. 6.

Comparison	Apex	Base
Intrapleural pressure Ppl (PIP)	More negative.	Less negative.
Transpulmonary pressure PTP	More positive.	Less positive.
Alveolar expansion at end-expiration	Already more expanded.	Less expanded.
Local compliance: Δvolume/Δpressure	Lower on the local pressure–volume curve.	Higher; more ready to expand.
Volume increase during inspiration	Smaller → less fresh ventilation.	Larger → more fresh ventilation.

Why does a bigger resting alveolus get less inspired air?

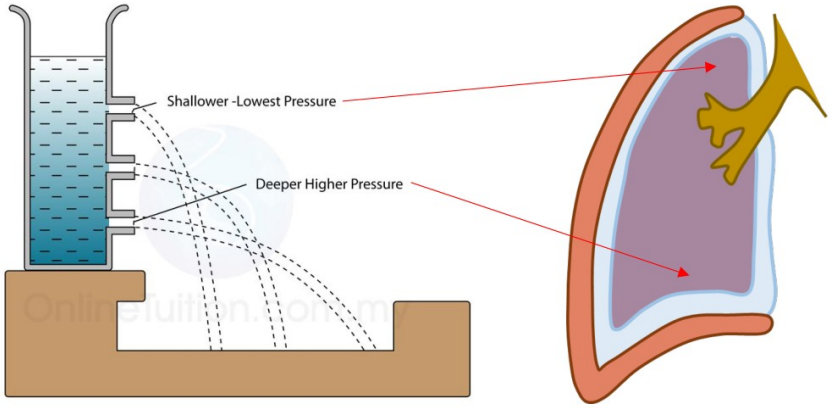
Gravity creates a pleural-pressure gradient. Apical alveoli start more stretched, so the same inspiratory pressure change produces a smaller volume increase.

Resting size and the extra volume entering during a breath are different quantities.

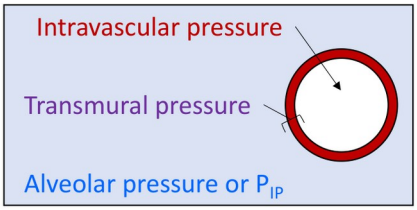
PTP = alveolar pressure − pleural pressure. At end-expiration, alveolar pressure is approximately atmospheric throughout a healthy upright lung; the regional pleural-pressure difference therefore matters. This describes regional mechanics and does not prove Tutorial Q1's word "always".

04 / Gravity changes perfusion and vessel calibre

Lecture PDF pp. 7–8 | Low pulmonary pressure: learning-outcome supplement



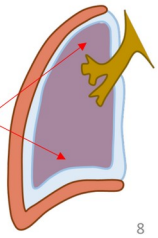
Original hydrostatic-pressure demonstration and lung diagram, lecture p. 7.



Transmural pressure = P_{inside} - $P_{outside}$

Intravascular P is high, transmural P is **high**. Compression/distension?

Intravascular P is low, transmural P is **low**. Compression/distension?



8

Original transmural-pressure diagram and selected distension/compression answers, lecture p. 8.

At comparable external pressure	Vessel behaviour	Effect on regional flow
Higher intravascular pressure	Higher transmural pressure → greater distension.	Lower vascular resistance → more flow.
Lower intravascular pressure	Lower transmural pressure → greater compression.	Higher resistance → less flow.

Transmural pressure = pressure inside − pressure outside the vessel. The relevant outside pressure depends on vessel location: alveolar pressure for alveolar capillaries; surrounding tissue pressure for extra-alveolar vessels. These pressures are not interchangeable in every vessel.

Why is pulmonary blood pressure low? [Learning-outcome supplement]

The lungs receive the same cardiac output as the systemic circuit, but pulmonary vascular resistance is much lower.

$\Delta P = \dot{Q} \times$ resistance. A compliant, distensible vascular network with many parallel pathways supports the flow at low pressure; recruitment/distension can accommodate increased flow.

Typical supplemental reference: mean pulmonary arterial pressure ~15 mmHg. This is not required by the slide diagrams.

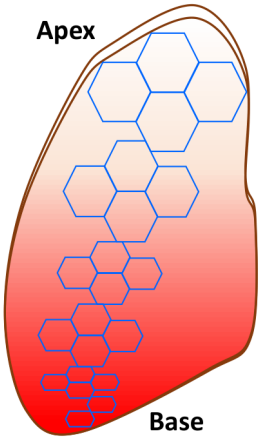
05 / Put V and Q together: apex versus base

Lecture PDF p. 9 | Tutorial Q2 | Supine position: learning-outcome supplement

Ventilation

More negative P_{IP}, More positive P_{TP}
More expanded alveoli, Less compliant (harder to expand an already expanded alveolus)
Less ventilation during inhalation

Less negative P_{IP}, Less positive P_{TP}
Less expanded alveoli, More compliant (more ready to expand)
More ventilation during inhalation



Perfusion

Lower intravascular pressure, so
Higher resistance
Less blood flow

Higher intravascular pressure, so
Lower resistance
More blood flow

Original completed regional ventilation/perfusion exercise, lecture p. 9.

Upright comparison	Apex	Base
Ventilation $\dot{V}A$	Lower.	Higher.
Perfusion $\dot{Q}$	Much lower.	Much higher.
V/Q ratio	Higher: Q falls more steeply than V.	Lower: Q increases more steeply than V.
Local alveolar / end-capillary gases	Relatively higher PO_2, lower PCO_2.	Relatively lower PO_2, higher PCO_2.

Tutorial Q2 · Original tutorial question

Blood leaving the pulmonary capillaries at the apex of an upright lung has a lower PO_2 than blood leaving the pulmonary capillaries at the base.

False | Compare the ratio, not ventilation alone

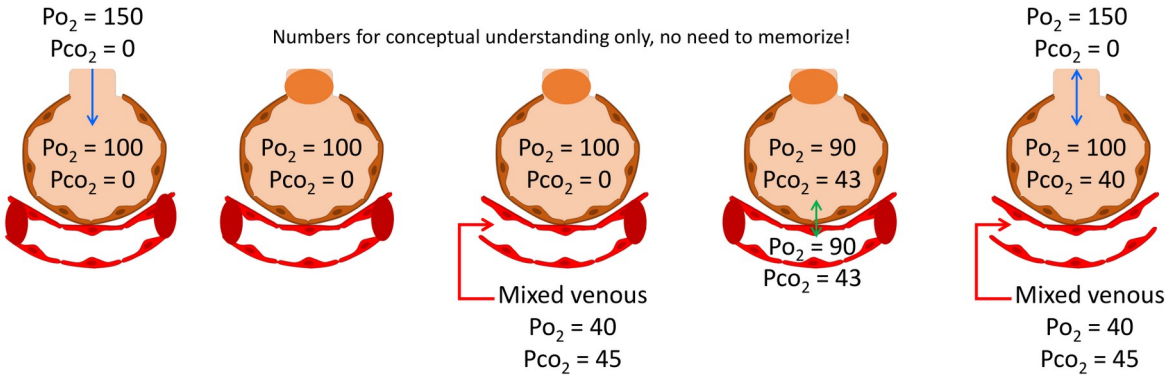
The apex receives less air, but an even greater relative reduction in blood flow → higher V/Q → higher local alveolar PO_2.
With adequate equilibration, blood leaving those capillaries has higher PO_2 than blood leaving basal capillaries. Do not confuse local end-capillary blood with mixed arterial blood.

What changes when supine? [Learning-outcome supplement]

The gravitational dependent direction becomes posterior/dorsal. Apex-to-base hydrostatic differences are smaller; the main gravity-related gradient is now back-to-front. Dependent regions generally receive more perfusion and often more ventilation during healthy spontaneous breathing.
Regional gradients are redistributed; do not copy upright apex/base rankings unchanged. Lung volume and airway closure can worsen dependent low V/Q, so posture alone does not guarantee perfect matching.

06 / V/Q determines alveolar gas composition

Lecture PDF pp. 12–13 | Source illustrations retained; intermediate cartoon numbers are conceptual

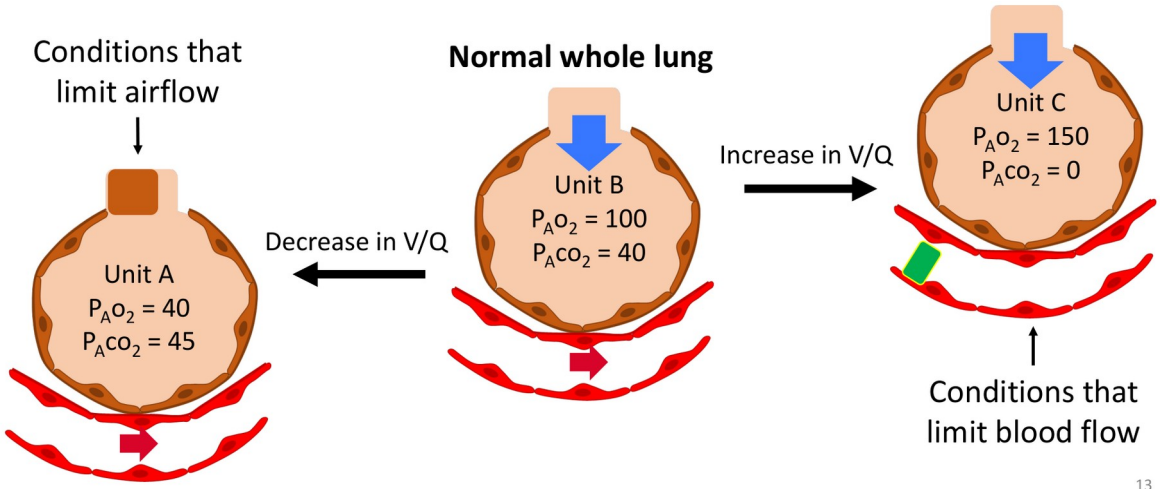


Common misconception: blood takes up oxygen, so P_Ao_2 must drop
No. Continuous ventilation maintains the P_Ao_2 and P_Aco_2 to be 100 and 40 mmHg respectively, despite continuous perfusion.

Original sequential alveolar-ventilation illustration, lecture p. 12.

Continuous exchange can have stable gas pressures

Blood continuously removes O_2 and supplies CO_2, while ventilation continuously replenishes O_2 and removes CO_2. At a matched steady state, PAO_2 and $PACO_2$ can stay approximately constant.
The temporary 90/43 and CO_2-free panels illustrate a changing scenario; they are not normal steady-state values to memorise.



Original low-, normal- and high-V/Q units, lecture p. 13.

V/Q limit	Alveolar gas tends towards...	Illustrative PO_2 / PCO_2, mmHg
0: perfused, not ventilated	Mixed venous blood composition.	~40 / 45.
Usual matched unit	Normal alveolar composition.	~100 / 40.
Infinity: ventilated, not perfused	Humidified inspired gas composition.	~150 / 0 on room air at sea level.

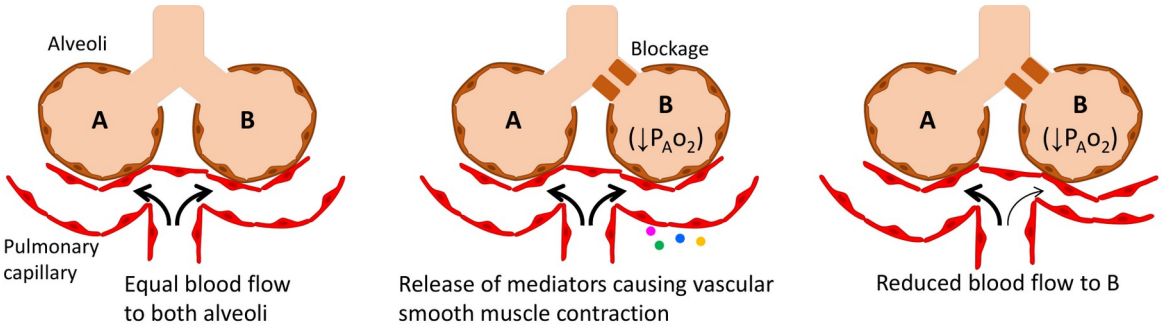
These are limiting examples, not fixed values for every low/high V/Q unit. Inspired pressure changes with altitude and FiO_2. A completely unperfused unit contributes no blood to the arterial mixture. Mix blood by flow and O_2 content, not a simple average of regional PO_2.

07 / Local matching: hypoxic vasoconstriction

Lecture PDF p. 10 | Blood-mixing calculation is an extension drawing on Lecture 2

only happen in lungs, hypoxia vasodialiation happen in rest of tissue

- Concept: in a normal upright lung, there is a mismatch between V and Q
  - Less blood goes to the apex, oxygen stays in the alveoli
  - Alveolar hypoxia stimulates vasoconstriction so as to divert blood to better-ventilated alveoli for gas exchange, minimizing V/Q mismatch



  - Elsewhere in the body, hypoxia causes vasodilation to deliver more blood

Original hypoxic vasoconstriction sequence and visible annotation, lecture p. 10.

Local pulmonary response

Airflow to unit B ↓ → PAO_2 in B ↓ → nearby pulmonary arterioles constrict
→ less blood goes to B → relatively more goes to better-ventilated units.
This helps matching; it does not remove the airway obstruction or guarantee full correction.

Where / how much hypoxia?	Expected response / significance
Local alveolar hypoxia in lung	Pulmonary vasoconstriction redistributes perfusion away from poorly ventilated alveoli.
Hypoxia in most systemic tissues	Local vasodilation generally increases O_2 delivery; opposite to the pulmonary response.
Widespread alveolar hypoxia [Extension]	Generalised pulmonary constriction can raise pulmonary vascular resistance/pressure; few better-oxygenated units may be available.

Integrated practice 1 | Self-written; not an original exam question

Two lung regions receive equal blood flow and produce end-capillary O_2 contents of 12 and 20 mL/dL. Calculate mixed content and delivery at total flow 5 L/min. Can raising PO_2 still further in the well-ventilated region fully compensate?
Reasoning: Mixed content = (12 + 20)/2 = 16 mL/dL; DO_2 = 16 × 5 × 10 = 800 mL/min (versus 1,000 if both were 20). Near-saturated Hb has little spare loading capacity; increasing high-unit PO_2 mainly adds a small amount of dissolved O_2. Mix oxygen content weighted by blood flow, not PO_2. Mismatch can cause hypoxemia without mandatory hypercapnia if other units eliminate extra CO_2.

08 / Alveolar ventilation versus minute ventilation

Lecture PDF p. 4 provides alveolar V; dead-space equations are extensions | Tutorial Q14, Q7

Term / symbol	Meaning / relationship
Tidal volume VT	Total volume of one breath.
Dead-space volume VD	Volume per breath that does not participate in effective gas exchange.
Frequency f	Breaths per minute.
Minute ventilation $\dot{V}E$	VT × f: total air moved per minute.
Alveolar ventilation $\dot{V}A$	(VT − VD) × f: effective fresh ventilation per minute in the simplified model.

Tutorial Q14 · Original tutorial question

Slow deep breathing and rapid shallow breathing provide the same alveolar ventilation if minute ventilation is identical.

False | Dead-space ventilation must be subtracted

Assume VD = 150 mL per breath and similar effective dead-space conditions. More small breaths waste a larger fraction of total ventilation in dead space.

Example	VT / f	$\dot{V}E$	$\dot{V}A$
Slower, deeper	500 mL / 12 min^{-1}	6.0 L/min	(500 − 150) × 12 = 4.2 L/min
Faster, shallower	250 mL / 24 min^{-1}	6.0 L/min	(250 − 150) × 24 = 2.4 L/min

Higher frequency alone does not establish higher alveolar ventilation. Physiological dead space can also increase when ventilated units lose perfusion.

Tutorial Q7 · Original tutorial question

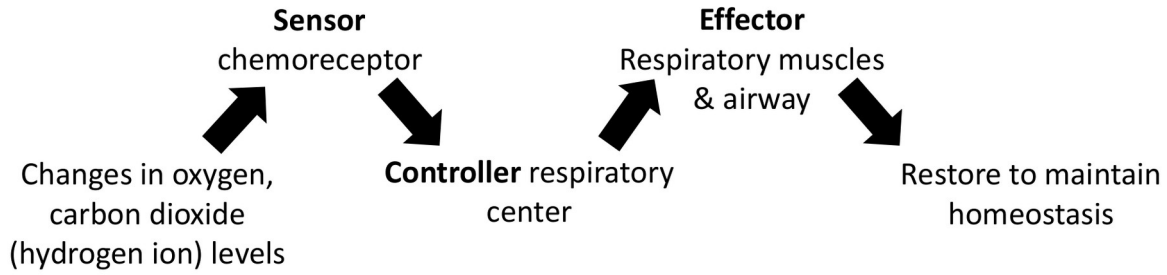
Hyperventilation can increase arterial PO_2 even when oxygen concentration in inspired air remains unchanged.

True under usual gas-exchange conditions

At unchanged FiO_2, increasing alveolar ventilation relative to CO_2 production lowers $PaCO_2$.
[Extension] $PAO_2 \approx PIO_2 - PaCO_2/R$ (R ~0.8) → PAO_2 rises → PaO_2 can rise.
"FiO_2 unchanged" is a condition, not the cause of increased ventilation. Shunt or other severe exchange impairment can limit the oxygenation response.

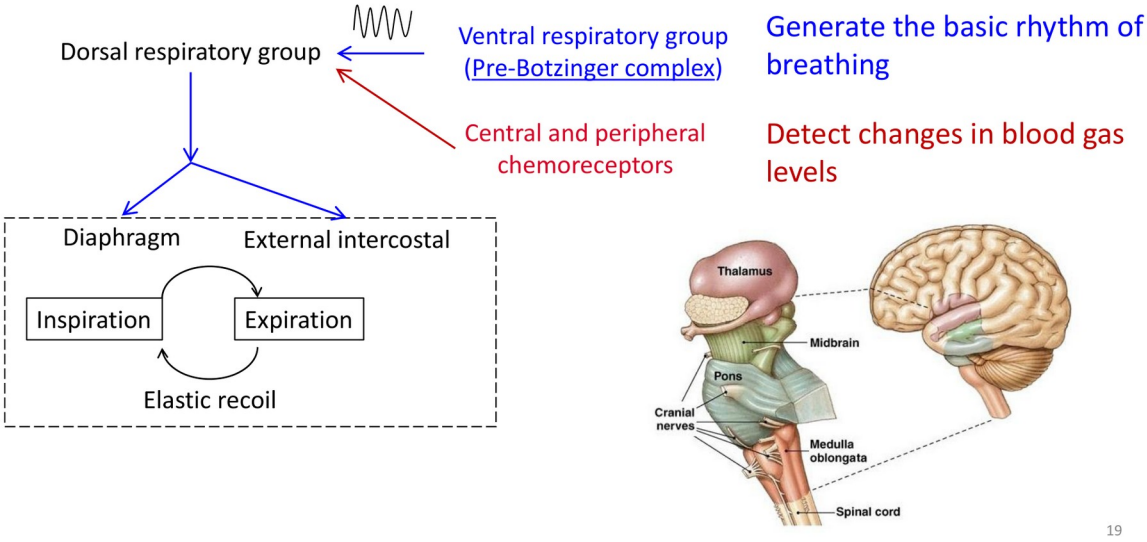
09 / Controller: rhythm and motor output

Lecture PDF pp. 16, 19, 21



Original sensor–controller–effector model, lecture p. 16.

Component	Role
Sensors: chemoreceptors	Detect chemical changes; send signals to the brainstem.
Controller: brainstem circuits	Integrate sensory/higher-centre input and organise respiratory rhythm and motor activity.
Effectors: muscles and airway	Change airflow and alveolar ventilation → alter blood gases → feedback to sensors.



Original brainstem diagram, muscle output and elastic recoil, lecture p. 19.

Named structure / action	Main point in this lecture
Pre-Bötzinger complex within VRG	Important rhythm-generating network in the ventral medulla; not a cardiac muscle pacemaker.
DRG: dorsal respiratory group	Integrates sensory input and contributes to inspiratory output in the lecture model.
Pons and medulla	Both participate in respiratory control; pontine circuits help shape the breathing pattern.
Quiet inspiration / expiration	Diaphragm and external intercostals contract / mainly passive elastic recoil. Forced expiration recruits muscles [Extension].

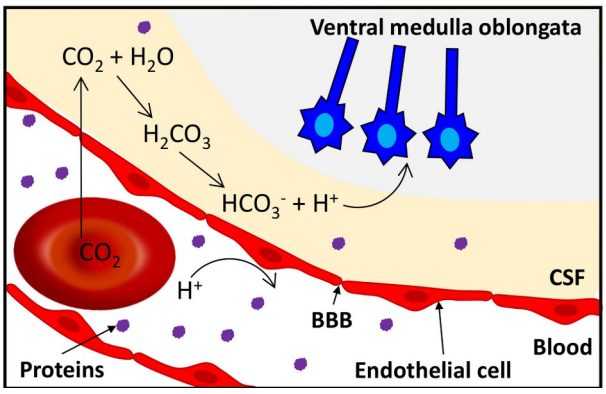
VRG = ventral respiratory group. The diagram is a simplified functional model of a distributed network. Slide p. 16 compares rhythm generation with the cardiac SA node and GI interstitial cells of Cajal; respiratory muscles themselves are driven by neural output.

10 / Central chemoreceptors and chronic CO_2

Lecture PDF p. 17 | Tutorial Q11 | Adaptation and extreme hypercapnia are extensions

what if too high CO2?

Central chemoreceptors (minute-to-minute control)
- Detect changes in H^+ the cerebrospinal fluid (CSF)
- Medulla of brainstem
  - Bath in CSF and the respiratory center is in the medulla



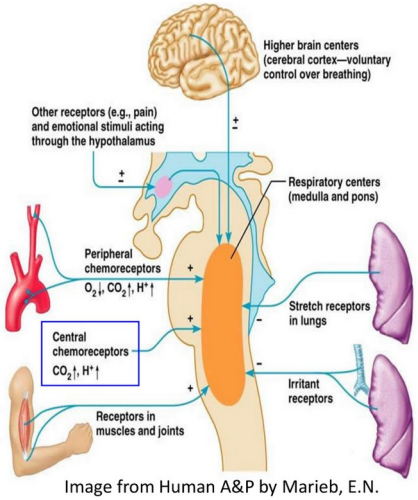


Image from Human A&P by Marieb, E.N.

Original central chemoreceptor/BBB diagram and annotated control overview, lecture p. 17; Human A&P by Marieb.

What do central receptors actually sense?

$PaCO_2 \uparrow \rightarrow CO_2$ crosses the blood–brain barrier (BBB) → enters cerebrospinal fluid (CSF) → $CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$ → CSF $H^+ \uparrow$ → medullary chemoreceptor stimulation → respiratory motor drive ↑ → alveolar ventilation ↑ → $PaCO_2$ falls: negative feedback.

- Central chemoreceptors respond to H^+ in the brain extracellular fluid/CSF. CO_2 is an effective arterial signal because it crosses the BBB readily; blood H^+ and HCO_3^- do not cross as freely.
- Central pathways supply much of the normal CO_2-driven, minute-to-minute feedback. Ventilatory drive means the neural stimulation that increases breathing.

Tutorial Q11 · Original tutorial question

A person with chronic hypercapnia depends more on arterial PO_2 than arterial PCO_2 as a ventilatory stimulus.

Qualified True in the usual introductory chronic-adaptation model

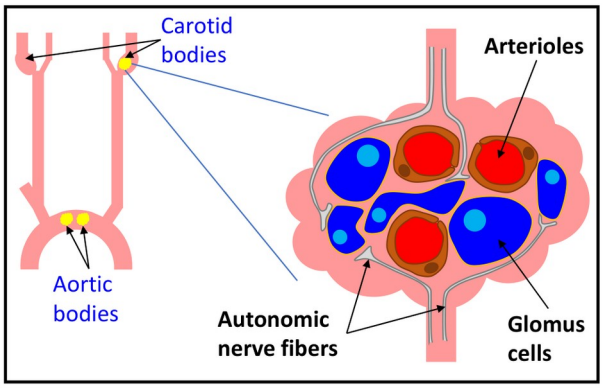
[Extension] Sustained $PaCO_2 \uparrow \rightarrow$ renal HCO_3^- retention and CSF buffering adaptation → CSF pH moves back towards normal → central CO_2/H^+ drive is attenuated.
Peripheral hypoxic stimulation can therefore contribute relatively more. This does not mean CO_2 responsiveness disappears or that every chronically hypercapnic person has an identical response.

Answer to the handwritten "what if too high CO_2?": an acute moderate rise usually stimulates ventilation; very severe hypercapnia can depress CNS function. Do not extrapolate an indefinitely increasing response. CSF = cerebrospinal fluid; CNS = central nervous system.

11 / Peripheral sensors: pressure, not O_2 content

Lecture PDF p. 18 | Exercise discussion and receptor comparison; detailed mechanisms are extensions

Peripheral chemoreceptors (in hypoxemia)
- Detect changes in arterial Po_2 mainly, also detect pH & Pco_2
- Bifurcation of common carotid arteries, along aortic arch



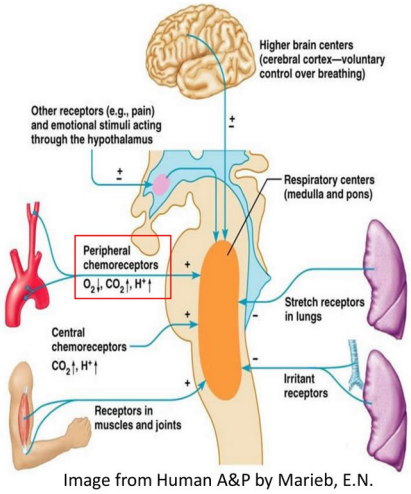


Image from Human A&P by Marieb, E.N.

Po2 and pCO2 remain stable even during exercise, why?

Original carotid/aortic body and glomus-cell diagram with annotations, lecture p. 18; overview from Marieb.

Compare	Central receptors	Peripheral receptors
Location	Medullary region; brain extracellular fluid/CSF.	Carotid bodies at carotid bifurcations; aortic bodies along the aortic arch.
Primary signals	CSF H^+, strongly influenced by arterial CO_2.	Arterial PO_2, plus PCO_2 and pH; carotid bodies are especially important for the arterial pH response.
Hypoxemia	No direct main O_2-sensing role in this model.	Low PaO_2 stimulates ventilation; response becomes much stronger below ~60 mmHg [Extension].
Blood metabolic acidosis	Blood H^+ does not rapidly act directly across the BBB.	Can stimulate ventilation even with normal PaO_2.

Glomus cells are specialised sensory cells in the peripheral chemoreceptor bodies; chemical changes alter the signals sent towards the brainstem.

Integrated practice 2 | Self-written; not an original exam question

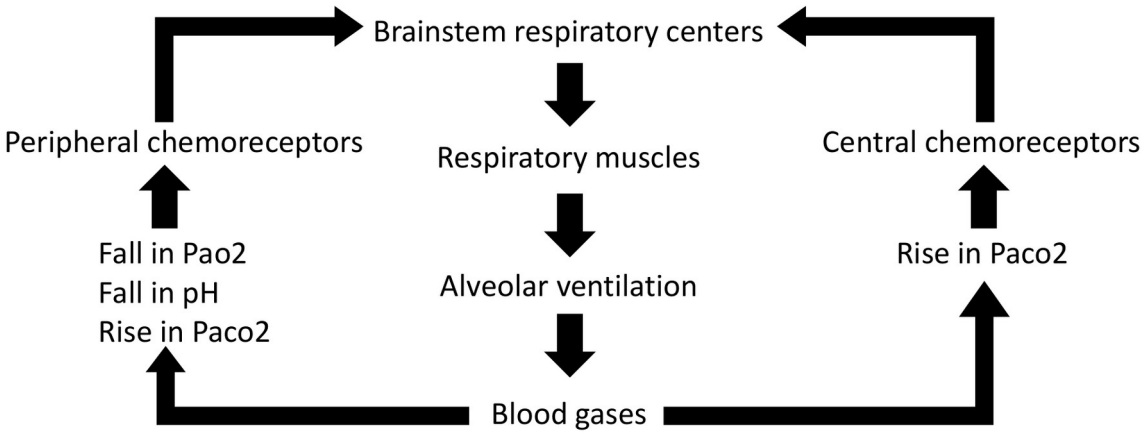
Compare A: severe anemia with normal PaO_2; B: PaO_2 50 mmHg with normal Hb; C: metabolic acidosis with normal PaO_2 and a compensatory fall in $PaCO_2$. Which signals drive ventilation, and why is "low O_2 content" not enough to predict the hypoxic chemoreflex?

Reasoning: A has reduced O_2 content but no strong low-PaO_2 signal solely from anemia. B strongly activates peripheral hypoxic sensing. C stimulates peripheral receptors through arterial H^+; the fall in $PaCO_2$ can reduce central CO_2 drive. Distinguish arterial pressure, content and the compartment containing H^+.

Slide exercise prompt: PaO_2/$PaCO_2$ often remain near normal during moderate exercise because ventilation rises with metabolic demand. [Extension] Feedforward motor signals and muscle/joint input help increase ventilation; stable gases do not mean breathing is unchanged.

12 / Chemoreflex applications: altitude and CA inhibition

Lecture PDF p. 20 | Tutorial Q4, Q12 | Alveolar gas/renal mechanisms are extensions



Original chemical feedback loop, lecture p. 20.

Tutorial Q4 · Original tutorial question

Hyperventilation at high altitude helps improve oxygenation even though it causes respiratory acidosis.

False | The acid–base label is wrong

Altitude: PB ↓ → inspired/alveolar PO_2 ↓ → peripheral hypoxic stimulation → ventilation ↑.
Acute hyperventilation → $PaCO_2$ ↓ → pH ↑: respiratory alkalosis, not acidosis. Reduced central CO_2/H^+ drive opposes the hypoxic stimulus.
[Extension] Over time, bicarbonate loss/buffering adaptation reduces the alkalotic brake and permits sustained increased ventilation.

Tutorial Q12 · Original tutorial question

Carbonic anhydrase inhibitor stimulates hyperventilation by promoting bicarbonate ion excretion.

True | Link a renal change to a respiratory response [Extension]

Acetazolamide → renal tubular carbonic anhydrase (CA) inhibition → HCO_3^- reabsorption ↓
→ urinary HCO_3^- loss ↑ → metabolic acidosis → chiefly peripheral pH stimulation → ventilation ↑.
At altitude this can reduce the inhibition caused by respiratory alkalosis. The primary mechanism here is renal base loss.

Why does inhibiting a reversible enzyme reduce bicarbonate recovery?

CA accelerates both directions at different sites: filtered HCO_3^- is converted to CO_2 in the tubular lumen; inside the cell, CO_2 is converted back to HCO_3^- for return to blood. Inhibition slows this recovery route.
H^+ used in this proximal recovery cycle is largely recycled, not necessarily excreted as net acid. Reversible catalysis does not mean the enzyme is unnecessary.

PB = barometric pressure; FiO_2 = inspired oxygen fraction; HCO_3^- = bicarbonate, an important base. Kidney/acid–base details support the original tutorial but are not fully taught in this slide deck.

13 / Protective reflexes and the breath-hold prompt

Protective reflexes: learning outcome on lecture p. 2; discussion prompt on p. 20

Protective reflexes [Learning-outcome supplement]

Named in the learning outcomes but not developed as a separate slide. The table supplies the standard mechanisms needed to address that outcome; it is not a verbatim lecturer-provided list.

Trigger / sensor	Reflex / response	Purpose / qualification
Irritation in larynx, trachea or bronchi	Cough: coordinated inspiration, compression and forceful expiration.	Helps clear irritants, secretions or foreign material from the airway.
Irritation of nasal mucosa	Sneeze.	Expels irritants from the upper airway.
Large lung inflation; pulmonary stretch receptors	Hering–Breuer inflation reflex: vagal input inhibits further inspiration.	Limits excessive inflation; more prominent at large tidal volumes and in infants than ordinary quiet adult breathing.
Laryngeal contact/irritation	Protective glottic closure, sometimes with transient apnea.	Helps prevent material entering the lower airway.

Other inputs visible in the original overview

- Cerebral cortex: voluntary control; hypothalamic pathways: emotion and pain-related influences.
- Muscle/joint receptors: contribute to the ventilatory response during activity.
- Protective airway/stretch signals modify the rhythm; chemical blood-gas feedback is not the only input.

Original classroom discussion: interpret the physiology

"How would you increase your chance of winning a breath-holding competition using your respiratory physiology knowledge?"

Explain the mechanism without treating it as a safe technique

During a breath hold, CO_2 rises and O_2 falls. Rising CO_2/CSF H^+ contributes strongly to the urge to breathe; falling PaO_2 increasingly activates peripheral receptors.

Pre-hold hyperventilation lowers starting $PaCO_2$ and can delay that urge, while adding little Hb-bound O_2 when Hb is already near saturation.

The delay can allow dangerous hypoxemia and loss of consciousness, especially in water. A delayed urge is not evidence of adequate oxygen; do not use hyperventilation to prolong breath holding.

Coverage check: matching and V/Q extremes pp. 2, 6–7; gravity and posture pp. 3–5; low pulmonary pressure p. 4; ventilation/dead space p. 8; rhythm and motor output p. 9; chemoreceptors pp. 10–12; protective reflexes p. 13. Only two self-written integrated practice problems are included.